INCH-POUND

A–A–59590/4B
7 April 2006
SUPERSEDING
A-A-59590/4A
21 June 2002

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/04	*	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Options

Example: AA59590/04EV28W is the PIN for a nickel finished, 2.8 inch (71.1 mm) long card holder. The card holder also features two tapped mounting holes for use with 2–56 UNC 2B fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

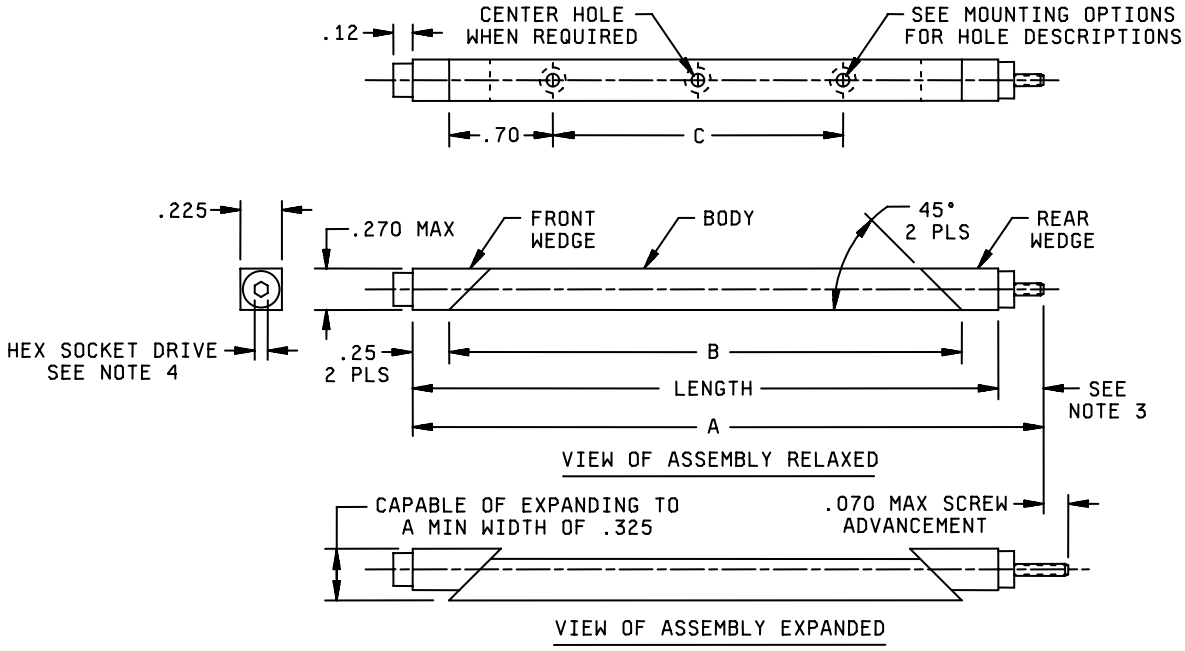
Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G–rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, in table I, and A–A–59590.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.070	1.8	.20	5.1	.250	6.4	.325	8.3
.12	3.1	.225	5.7	.270	6.9	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.5 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders without option “W” or “D” and .30 (7.6 mm) maximum on card holders with option “W” or “D”.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (.098 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ± .02 (0.5)	Dimension "A" +.02 / –.09 (+0.5 / –2.3)		Dimension "B" ± .03 (0.8)	Dimension "C" ± .005 (0.13)
		Without options W or D	With options W or D		
18	1.8 (45.7)	2.00 (50.8)	2.10 (53.3)	1.30 (33.0)	.75 (19.1)
23	2.3 (58.4)	2.50 (63.5)	2.60 (66.0)	1.80 (45.7)	.40 (10.2)
28	2.8 (71.1)	3.00 (76.2)	3.10 (78.7)	2.30 (58.4)	.90 (22.9)
30	3.0 (76.2)	3.20 (81.3)	3.30 (83.8)	2.50 (63.5)	1.10 (27.9)
33	3.3 (83.8)	3.50 (88.9)	3.60 (91.4)	2.80 (71.1)	1.40 (35.6)
35	3.5 (88.9)	3.70 (94.0)	3.80 (96.5)	3.00 (76.2)	1.60 (40.6)
38	3.8 (96.5)	4.00 (101.6)	4.10 (104.1)	3.30 (83.8)	1.90 (48.3)
43	4.3 (109.2)	4.50 (114.3)	4.60 (116.8)	3.80 (96.5)	2.40 (61.0)
48	4.8 (121.9)	5.00 (127.0)	5.10 (129.1)	4.30 (109.2)	2.90 (73.7)
53	5.3 (134.6)	5.50 (139.7)	5.60 (142.2)	4.80 (121.9)	3.40 (86.4)
55	5.5 (139.7)	5.70 (144.8)	5.80 (147.3)	5.00 (127.0)	3.60 (91.4)
58	5.8 (147.3)	6.00 (152.4)	6.10 (154.9)	5.30 (134.6)	3.90 (99.1)
61	6.1 (154.9)	6.30 (160.0)	6.40 (162.6)	5.60 (142.2)	4.20 (106.7)
63	6.3 (160.0)	6.50 (165.1)	6.60 (167.6)	5.80 (147.3)	4.40 (111.8)
67	6.7 (170.2)	6.90 (175.3)	7.00 (177.8)	6.20 (157.5)	4.80 (121.9)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Finish. The finish options shall be as specified in A–A–59590. The finish options available for this specification sheet are as follows: "B" (black anodize), "C" (chemical film), "E" (electroless nickel), "H" (hard black anodize), and "P" (electrodeposited nickel).

Mounting options. The mounting options shall be as specified in A–A–59590. The mounting options available for this specification sheet are as follows: "J" or "K" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0-80 UNF 2B holes), "V" or "S" (tapped 2-56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" or "K" may be shipped unassembled.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

Options. Card holders can have options “W” (lockwasher and flat washer), “E” (screw self-locking element), or “D” (combination of options “W” and “E”) as described below. Card holders not requiring these options shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 2). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

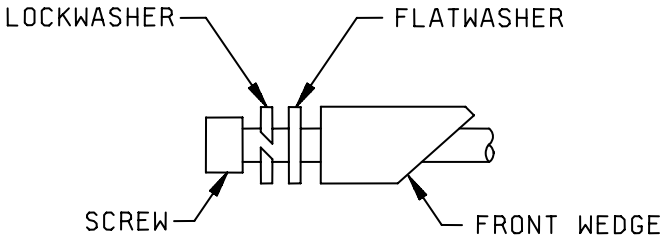


FIGURE 2. Lockwasher and flat washer details.

Screw self-locking element (see figure 3). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

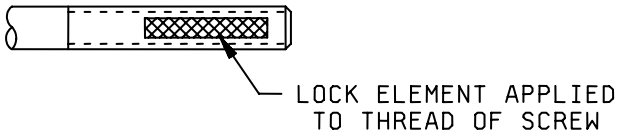


FIGURE 3. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher Pentair Electronic Packaging 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: infobirtcher@pentair-ep.com URL: www.birtcherproducts.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers in table II and on DSCC Drawing 84103 (table III) as shown. The CID PINs listed in table II are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	40–6	CJ28	225CR2.80H
CN28	40–6–A	CN28	225CR2.80
CT28	40–6–T	CT28	225CR2.80T0
CV28	40–6–S	CV28	225CR2.80T2
CG28	40–6–M	CG28	M225CR2.80TM2.5
CJ28E	40–6–L	CJ28E	225CR2.80HL
CN28E	40–6–A–L	CN28E	225CR2.80L
CT28E	40–6–T–L	CT28E	225CR2.80T0L
CV28E	40–6–S–L	CV28E	225CR2.80T2L
CG28E	40–6–M–L	CG28E	M225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094
CJ28W	40–6–LF	CJ28W	W225CR2.80H
CN28W	40–6–LF–A	CN28W	W225CR2.80
CT28W	40–6–LF–T	CT28W	W225CR2.80T0
CV28W	40–6–LF–S	CV28W	W225CR2.80T2
CG28W	40–6–LF–M	CG28W	MW225CR2.80TM2.5
CJ28D	40–6–LF–L	CJ28D	W225CR2.80H
CN28D	40–6–LF–A–L	CN28D	W225CR2.80
CT28D	40–6–LF–T–L	CT28D	W225CR2.80T0
CV28D	40–6–LF–S–L	CV28D	W225CR2.80T2
CG28D	40–6–LF–M–L	CG28D	MW225CR2.80TM2.5
BJ28	40–6–B	BJ28	A225CR2.80H
BN28	40–6–B–A	BN28	A225CR2.80
BT28	40–6–B–T	BT28	A225CR2.80T0
BV28	40–6–B–S	BV28	A225CR2.80T2
BG28	40–6–B–M	BG28	MA225CR2.80TM2.5
BJ28E	40–6–B–L	BJ28E	A225CR2.80HL
BN28E	40–6–B–A–L	BN28E	A225CR2.80L
BT28E	40–6–B–T–L	BT28E	A225CR2.80T0L
BV28E	40–6–B–S–L	BV28E	A225CR2.80T2L
BG28E	40–6–B–M–L	BG28E	MA225CR2.80TM2.5L
BJ28W	40–6–B–LF	BJ28W	WA225CR2.80H
BN28W	40–6–B–LF–A	BN28W	WA225CR2.80
BT28W	40–6–B–LF–T	BT28W	WA225CR2.80T0
BV28W	40–6–B–LF–S	BV28W	WA225CR2.80T2
BG28W	40–6–B–LF–M	BG28W	MWA225CR2.80TM2.5
BJ28D	40–6–B–LF–L	BJ28D	WA225CR2.80HL
BN28D	40–6–B–LF–A–L	BN28D	WA225CR2.80L
BT28D	40–6–B–LF–T–L	BT28D	WA225CR2.80T0L
BV28D	40–6–B–LF–S–L	BV28D	WA225CR2.80T2L
BG28D	40–6–B–LF–M–L	BG28D	MWA225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094
HJ28	40–6–B3	HJ28	HA225CR2.80H
HN28	40–6–B3–A	HN28	HA225CR2.80
HT28	40–6–B3–T	HT28	HA225CR2.80T0
HV28	40–6–B3–S	HV28	HA225CR2.80T2
HG28	40–6–B3–M	HG28	MHA225CR2.80TM2.5
HJ28E	40–6–B3–L	HJ28E	HA225CR2.80HL
HN28E	40–6–B3–A–L	HN28E	HA225CR2.80L
HT28E	40–6–B3–T–L	HT28E	HA225CR2.80T0L
HV28E	40–6–B3–S–L	HV28E	HA225CR2.80T2L
HG28E	40–6–B3–M–L	HG28E	MHA225CR2.80TM2.5L
HJ28W	40–6–B3–LF	HJ28W	WHA225CR2.80H
HN28W	40–6–B3–LF–A	HN28W	WHA225CR2.80
HT28W	40–6–B3–LF–T	HT28W	WHA225CR2.80T0
HV28W	40–6–B3–LF–S	HV28W	WHA225CR2.80T2
HG28W	40–6–B3–LF–M	HG28W	MWHA225CR2.80TM2.5
HJ28D	40–6–B3–LF–L	HJ28D	WHA225CR2.80HL
HN28D	40–6–B3–LF–A–L	HN28D	WHA225CR2.80L
HT28D	40–6–B3–LF–T–L	HT28D	WHA225CR2.80T0L
HV28D	40–6–B3–LF–S–L	HV28D	WHA225CR2.80T2L
HG28D	40–6–B3–LF–M–L	HG28D	MWHA225CR2.80TM2.5L
PJ28	40–6–N	EJ28	EN225CR2.80H
PN28	40–6–N–A	EN28	EN225CR2.80
PT28	40–6–N–T	ET28	EN225CR2.80T0
PV28	40–6–N–S	EV28	EN225CR2.80T2
PG28	40–6–N–M	EG28	MEN225CR2.80TM2.5
PJ28E	40–6–N–L	EJ28E	EN225CR2.80HL
PN28E	40–6–N–A–L	EN28E	EN225CR2.80L
PT28E	40–6–N–T–L	ET28E	EN225CR2.80T0L
PV28E	40–6–N–S–L	EV28E	EN225CR2.80T2L
PG28E	40–6–N–M–L	EG28E	MEN225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/04	Vendor similar designator or type part number 1/ CAGE 52094
PJ28W	40–6–N–LF	EJ28W	WEN225CR2.80H
PN28W	40–6–N–LF–A	EN28W	WEN225CR2.80
PT28W	40–6–N–LF–T	ET28W	WEN225CR2.80T0
PV28W	40–6–N–LF–S	EV28W	WEN225CR2.80T2
PG28W	40–6–N–LF–M	EG28W	MWEN225CR2.80TM2.5
PJ28D	40–6–N–LF–L	EJ28D	WEN225CR2.80HL
PN28D	40–6–N–LF–A–L	EN28D	WEN225CR2.80L
PT28D	40–6–N–LF–T–L	ET28D	WEN225CR2.80T0L
PV28D	40–6–N–LF–S–L	EV28D	WEN225CR2.80T2L
PG28D	40–6–N–LF–M–L	EG28D	MWEN225CR2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Supersession information.

Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN
84103–01B	AA59590/04BJ28	84103–01C	AA59590/04CJ28
84103–02B	AA59590/04BJ30	84103–02C	AA59590/04CJ30
84103–03B	AA59590/04BJ35	84103–03C	AA59590/04CJ35
84103–04B	AA59590/04BJ38	84103–04C	AA59590/04CJ38
84103–05B	AA59590/04BJ48	84103–05C	AA59590/04CJ48
84103–06B	AA59590/04BJ53	84103–06C	AA59590/04CJ53
84103–07B	AA59590/04BJ58	84103–07C	AA59590/04CJ58

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/04	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/04	Vendor similar designator or type part number 1/ 2/ CAGE 52094
PV18W	40–4–N–LF–S	EV18W	WEN225CR1.80T2
PV23W	40–5–N–LF–S	EV23W	WEN225CR2.30T2
PV28W	40–6–N–LF–S	EV28W	WEN225CR2.80T2
PV30W	3/	EV30W	WEN225CR3.00T2
PV33W	40–7–N–LF–S	EV33W	WEN225CR3.30T2
PV35W	3/	EV35W	WEN225CR3.50T2
PV38W	40–8–N–LF–S	EV38W	WEN225CR3.80T2
PV43W	40–9–N–LF–S	EV43W	WEN225CR4.30T2
PV48W	40–10–N–LF–S	EV48W	WEN225CR4.80T2
PS53W	40–11–N–LF–S	ES53W	WEN225CR5.30ET2
PS55W	3/	ES55W	WEN225CR5.50ET2
PS58W	40–12–N–LF–S	ES58W	WEN225CR5.80ET2
PS61W	3/	ES61W	WEN225CR6.10ET2
PS63W	40–13–N–LF–S	ES63W	WEN225CR6.30ET2
PS67W	3/	ES67W	WEN225CR6.70ET2

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–013

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.